Circle Academic Resources

Business Mathematics

Examination Practice Handbook

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A Note to the Reader

Dear Reader,

Your examination preparation is incomplete without practice. Understanding concepts is important, but testing your knowledge helps you identify gaps, improve accuracy, and build confidence.

This Practice Handbook brings together important questions from all five courses prescribed for your BBA 1st Semester End-Term Examinations. These are questions we believe every student should attempt before entering the examination hall.

The handbook is designed to help you revise key concepts, assess your understanding, and become familiar with examination-oriented questions.

A special thank-you to our academic toppers for their valuable feedback and contributions in preparing these resources.

Practise with purpose. Learn from your mistakes. Enter the examination with confidence.

The Circle Practice Handbook

BUSINESS MATHEMATICS

BBA CIRCLE

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BUSINESS MATHEMATICS

50-QUESTION PRACTICE SET

25 Multiple Choice Questions · 25 Multiple Selection Questions

INSTRUCTIONS

Multiple Choice Questions: Select one correct option for each question.

Multiple Selection Questions: Select all correct options. A question may have more than one correct answer.

Multiple Selection Question marking:

1. One correct option + one wrong option selected: **No marks.**
2. Only one correct option selected, without selecting any incorrect option: **Partial marks.**
3. All correct options selected: **Full marks.**
4. Even if one incorrect option is selected: **Zero marks.**

SECTION A · MULTIPLE CHOICE QUESTIONS

Q01 · Multiple Choice Questions | 1 practice mark

A business can receive Rs. 2,00,000 today or the same amount one year later. Assuming a positive rate of return, why is receiving the money today generally preferable?

A. The future amount is automatically larger

B. The money received today can be invested or used immediately

C. The amount received today has no opportunity cost

D. The two amounts must always have identical value

Answer: (B)

Q02 · Multiple Choice Questions | 1 practice mark

A firm invests Rs. 80,000 at 9% simple interest per annum for 10 months. How much interest will it earn?

A. Rs. 5,400

B. Rs. 7,200

C. Rs. 6,000

D. Rs. 8,000

Answer: (C)

Q03 · Multiple Choice Questions | 1 practice mark

A business borrows Rs. 40,000 at 8% simple interest per annum for 9 months. What is the interest payable?

A. Rs. 3,200

B. Rs. 2,000

C. Rs. 1,800

D. Rs. 2,400

Answer: (D)

Q04 · Multiple Choice Questions | 1 practice mark

A deposit of Rs. 50,000 earns 10% simple interest per annum for 2 years. What is its maturity value?

A. Rs. 60,000

B. Rs. 55,000

C. Rs. 65,000

D. Rs. 70,000

Answer: (A)

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Q05 · Multiple Choice Questions | 1 practice mark

A payment of Rs. 66,000 is due after one year. If the relevant simple discount rate is 10% per annum, what is its present value under the simple-interest model?

A. Rs. 62,000

B. Rs. 60,000

C. Rs. 59,400

D. Rs. 56,000

Answer: (B)

Q06 · Multiple Choice Questions | 1 practice mark

A bill has a face value of Rs. 1,50,000 and is discounted at 8% simple bank discount for 9 months. What are the bank discount and proceeds?

A. Discount Rs. 12,000; proceeds Rs. 1,38,000

B. Discount Rs. 6,000; proceeds Rs. 1,44,000

C. Discount Rs. 9,000; proceeds Rs. 1,41,000

D. Discount Rs. 10,000; proceeds Rs. 1,40,000

Answer: (C)

Q07 · Multiple Choice Questions | 1 practice mark

A company invests Rs. 25,000 at 8% compound interest per annum for 2 years. What is the future value?

A. Rs. 27,000

B. Rs. 28,500

C. Rs. 30,000

D. Rs. 29,160

Answer: (D)

Q08 · Multiple Choice Questions | 1 practice mark

A bank quotes a nominal annual interest rate of 12%, compounded quarterly. What is the periodic quarterly rate?

A. 12%

B. 3%

C. 4%

D. 1%

Answer: (B)

Q09 · Multiple Choice Questions | 1 practice mark

A deposit earns a nominal annual rate of 10%, compounded monthly. What is the approximate effective annual rate?

A. 10.00%

B. 10.25%

C. 10.47%

D. 11.00%

Answer: (C)

Q10 · Multiple Choice Questions | 1 practice mark

What is the present value of Rs. 13,310 receivable after 3 years, discounted at 10% per annum compounded annually?

A. Rs. 10,000

B. Rs. 11,000

C. Rs. 12,100

D. Rs. 9,500

Answer: (A)

Q11 · Multiple Choice Questions | 1 practice mark

Using the Rule of 72, approximately how long will an investment take to double at 6% annual compound growth?

A. 6 years

B. 9 years

C. 10 years

D. 12 years

Answer: (D)

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Q12 · Multiple Choice Questions | 1 practice mark

An investment grows from Rs. 50,000 to Rs. 60,500 in 2 years with annual compounding. What is its compound annual growth rate?

A. 8%

B. 10%

C. 11%

D. 12%

Answer: (B)

Q13 · Multiple Choice Questions | 1 practice mark

A fund grows from Rs. 1,00,000 to Rs. 1,50,000 at 10% annual compound interest. Approximately how long does it take?

A. 4.25 years

B. 3 years

C. 5 years

D. 6 years

Answer: (A)

Q14 · Multiple Choice Questions | 1 practice mark

A business deposits Rs. 5,000 at the end of each year for 4 years in an account earning 8% annually. What is the approximate future value at the end of year 4?

A. Rs. 20,000

B. Rs. 22,000

C. Rs. 22,530.56

D. Rs. 24,000

Answer: (C)

Q15 · Multiple Choice Questions | 1 practice mark

Four annual payments of Rs. 5,000 are made at the beginning of each year. At 10% annual interest, what is their approximate value at the end of year 4?

A. Rs. 23,205

B. Rs. 25,525.50

C. Rs. 24,000

D. Rs. 20,000

Answer: (B)

Q16 · Multiple Choice Questions | 1 practice mark

A firm wants Rs. 1,21,000 after 2 years by making equal deposits at the end of each year. If the fund earns 10% annually, what annual deposit is required?

A. Rs. 60,500

B. Rs. 55,000

C. Rs. 50,000

D. Approximately Rs. 57,619

Answer: (D)

Q17 · Multiple Choice Questions | 1 practice mark

What is the present value of two annual payments of Rs. 12,000, each received at the end of the year, at a 10% annual discount rate?

A. Approximately Rs. 20,826.45

B. Rs. 24,000

C. Rs. 22,000

D. Rs. 18,000

Answer: (A)

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Q18 · Multiple Choice Questions | 1 practice mark

A business has an opening loan balance of Rs. 1,50,000 at 12% nominal annual interest compounded monthly. What is the interest portion of the first monthly EMI?

A. Rs. 1,200

B. Rs. 1,500

C. Rs. 1,800

D. Rs. 2,000

Answer: (B)

Q19 · Multiple Choice Questions | 1 practice mark

The first monthly EMI on the loan in the previous example is Rs. 5,000. How much of the first payment reduces the principal?

A. Rs. 5,000

B. Rs. 1,500

C. Rs. 3,500

D. Rs. 6,500

Answer: (C)

Q20 · Multiple Choice Questions | 1 practice mark

After the first EMI in the previous example, what is the approximate closing loan balance, assuming no fees or other adjustments?

A. Rs. 1,50,000

B. Rs. 1,48,500

C. Rs. 1,47,000

D. Rs. 1,46,500

Answer: (D)

Q21 · Multiple Choice Questions | 1 practice mark

Under a standard loan-amortisation model, how is the outstanding balance immediately after a payment generally determined?

A. By adding all past EMIs to the original loan

B. As the present value of the remaining EMIs at the balance date

C. By subtracting only the interest paid from the original loan

D. By dividing the original loan equally by the number of months

Answer: (B)

Q22 · Multiple Choice Questions | 1 practice mark

A business considers refinancing a loan because a new lender offers a lower monthly EMI. What should the business compare before deciding?

A. Only the new EMI amount

B. Only the advertised interest rate

C. The remaining term, total interest, fees, settlement amount, and cash-flow effect

D. Only the age of the original loan

Answer: (C)

Q23 · Multiple Choice Questions | 1 practice mark

A perpetuity pays Rs. 8,000 at the end of every year indefinitely. If the discount rate is 8% per annum, what is its value one period before the first payment?

A. Rs. 64,000

B. Rs. 80,000

C. Rs. 1,20,000

D. Rs. 1,00,000

Answer: (D)

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Q24 · Multiple Choice Questions | 1 practice mark

A firm's profit function is P(q) = -2q^2 + 80q - 500. At what output quantity does the quadratic function reach its maximum?

A. 20 units

B. 40 units

C. 80 units

D. 10 units

Answer: (A)

Q25 · Multiple Choice Questions | 1 practice mark

A break-even model produces the quadratic equation q^2 - 7q + 12 = 0. What are the two algebraic break-even quantities?

A. 1 and 12 units

B. 2 and 6 units

C. 3 and 4 units

D. 5 and 7 units

Answer: (C)

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SECTION B · MULTIPLE SELECTION QUESTIONS

Q26 · Multiple Selection Questions | 1 practice mark

Which statements explain why the time value of money matters in business decisions?

A. Cash received earlier may be invested or used to earn a return

B. Cash flows at different dates should be compared at a common valuation date

C. Equal nominal amounts always have equal economic value regardless of timing

D. Discounting converts a future amount into a comparable present amount

Answer: (A, B and D)

Q27 · Multiple Selection Questions | 1 practice mark

Which statements correctly describe a simple-interest calculation?

A. Interest is calculated on the original principal

B. If the annual rate is 9%, the rate is entered as 0.09 in the decimal form

C. Time may be entered in months while the annual rate is used without conversion

D. The time unit must be consistent with the stated rate period

Answer: (A, B and D)

Q28 · Multiple Selection Questions | 1 practice mark

Which relationships are correctly used in simple maturity value and bank-discount problems?

A. Simple maturity value: A = P(1 + rt)

B. Simple present value: P = A / (1 + rt)

C. Bank discount: D = Fdt, when d is the annual discount rate and t is in years

D. Bank-discount proceeds: proceeds = face value + discount

Answer: (A, B and C)

Q29 · Multiple Selection Questions | 1 practice mark

Which statements about compound interest are correct?

A. Interest can be earned on previously accumulated interest

B. For a single annual compounding period, FV = P(1 + i)^n

C. Compound interest is always lower than simple interest at the same positive rate and term

D. The rate per period and number of periods must match the compounding frequency

Answer: (A, B and D)

Q30 · Multiple Selection Questions | 1 practice mark

A bank quotes a nominal annual rate compounded several times per year. Which conversions are correct?

A. Quarterly compounding uses m = 4

B. Monthly compounding uses m = 12

C. The periodic rate is the nominal annual rate divided by m

D. For t years, the number of periods is n = mt

Answer: (A, B, C and D)

Q31 · Multiple Selection Questions | 1 practice mark

Which statements about the effective annual rate (EAR) are correct?

A. EAR = (1 + j/m)^m - 1

B. For a positive nominal rate, more frequent compounding generally increases EAR

C. EAR is always equal to the nominal annual rate divided by m

D. EAR expresses the total annual growth after allowing for within-year compounding

Answer: (A, B and D)

Q32 · Multiple Selection Questions | 1 practice mark

Which statements about compound growth, time, and the Rule of 72 are correct?

A. Compound time can be found using n = ln(FV/PV) / ln(1+i)

B. CAGR can be found using (FV/PV)^(1/n) - 1

C. Rule of 72 estimates doubling time as approximately 72 divided by the annual rate in percent

D. Rule of 72 gives an exact doubling period at every interest rate

Answer: (A, B and C)

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Q33 · Multiple Selection Questions | 1 practice mark

Which statements describe an ordinary annuity correctly?

A. Payments are made at the end of each period

B. It involves a series of regular payments

C. The payment amount and interval are defined for the annuity

D. The first payment is necessarily made at the beginning of the first period

Answer: (A, B and C)

Q34 · Multiple Selection Questions | 1 practice mark

Which statements about an annuity due are correct?

A. Payments are made at the beginning of each period

B. Its future value is the ordinary-annuity future value multiplied by (1+i), for the same payment count and rate

C. Its present value is the ordinary-annuity present value multiplied by (1+i), under matching timing assumptions

D. Payment timing has no effect on the value

Answer: (A, B and C)

Q35 · Multiple Selection Questions | 1 practice mark

A business is planning a sinking fund to build a target amount. Which statements are correct?

A. It uses periodic deposits to accumulate a specified future amount

B. The deposit calculation depends on the target, rate, and number of periods

C. For the same target and rate, a shorter saving period generally requires larger deposits

D. A sinking fund target is calculated by adding all deposits without considering interest

Answer: (A, B and C)

Q36 · Multiple Selection Questions | 1 practice mark

Which statements about the present value of an annuity are correct?

A. Each future payment is discounted to the selected valuation date

B. Payment timing affects present value

C. At a positive discount rate, a later equal payment has a lower present value than an earlier one

D. The undiscounted total of payments is always equal to present value

Answer: (A, B and C)

Q37 · Multiple Selection Questions | 1 practice mark

Which statements correctly describe an EMI and its amortisation?

A. An EMI includes an interest portion and a principal-repayment portion

B. Periodic interest is calculated using the opening balance and the applicable periodic rate

C. Principal repaid = EMI - interest for that period

D. The entire EMI is principal repayment from the first instalment

Answer: (A, B and C)

Q38 · Multiple Selection Questions | 1 practice mark

Which statements generally describe a reducing-balance loan with a fixed EMI and fixed rate?

A. The opening balance for the next period is the previous closing balance

B. As the balance falls, the interest portion generally falls

C. The principal portion generally rises if the EMI and rate remain unchanged

D. The principal portion is necessarily identical in every instalment

Answer: (A, B and C)

Q39 · Multiple Selection Questions | 1 practice mark

Which statements about an outstanding loan balance are correct?

A. Under standard assumptions, it can be found as the present value of remaining EMIs at the balance date

B. The timing of the balance measurement relative to an EMI matters

C. It is always equal to the sum of all EMIs originally scheduled

D. The outstanding principal is relevant when comparing payoff or refinancing choices

Answer: (A, B and D)

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Q40 · Multiple Selection Questions | 1 practice mark

Which factors should be considered when evaluating a refinancing offer?

A. Settlement or payoff amount on the existing loan

B. Fees and other transaction costs

C. New rate, instalment, and remaining term

D. Monthly EMI alone, without considering total cost

Answer: (A, B and C)

Q41 · Multiple Selection Questions | 1 practice mark

Which statements about a perpetuity are correct?

A. It represents a payment stream continuing indefinitely

B. For a constant payment made each period, PV = PMT/i under the stated timing convention

C. A higher positive discount rate lowers the value of the same constant perpetuity payment

D. A perpetuity has a fixed final payment date

Answer: (A, B and C)

Q42 · Multiple Selection Questions | 1 practice mark

Which relationships are useful in analysing revenue, quantity, and linear cost?

A. Total revenue can be expressed as price multiplied by quantity

B. A linear cost model may be written as fixed cost plus variable cost per unit times quantity

C. Capacity constraints can limit feasible output even when demand exists

D. A price reduction always increases revenue and profit

Answer: (A, B and C)

Q43 · Multiple Selection Questions | 1 practice mark

Which statements about linear cost and marginal output are correct?

A. In a linear cost model, the slope represents the variable cost per additional unit

B. Fixed cost is the cost represented by the intercept when output is zero

C. Marginal output measures the additional output from an additional unit of input

D. A linear cost model means that total revenue must also be linear in every market

Answer: (A, B and C)

Q44 · Multiple Selection Questions | 1 practice mark

Which statements about a quadratic function are correct?

A. If the coefficient of q^2 is negative, the parabola opens downward

B. The vertex occurs at q = -b/(2a)

C. If the coefficient of q^2 is positive, the vertex is a minimum

D. The sign of a has no effect on whether the vertex is a maximum or minimum

Answer: (A, B and C)

Q45 · Multiple Selection Questions | 1 practice mark

Which statements about roots and break-even analysis are correct?

A. Break-even quantities can be found by solving the profit equation P(q) = 0

B. The discriminant b^2 - 4ac helps identify the number of real roots

C. An algebraic root must still be checked against feasible business quantities

D. Every algebraic root is automatically feasible in the business context

Answer: (A, B and C)

Q46 · Multiple Selection Questions | 1 practice mark

Which statements are important when building a quadratic demand or revenue model?

A. Identify whether price is expressed as a function of quantity or quantity as a function of price

B. Revenue is calculated as price multiplied by quantity

C. Maximum revenue and maximum profit need not occur at the same quantity or price

D. The maximum-revenue choice always maximises profit

Answer: (A, B and C)

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Q47 · Multiple Selection Questions | 1 practice mark

Which checks help interpret a business pricing model responsibly?

A. Identify whether the objective is revenue, profit, or another outcome

B. Consider demand response and feasible quantity

C. Check capacity and relevant cost behaviour

D. Assume the highest listed price is always the best decision

Answer: (A, B and C)

Q48 · Multiple Selection Questions | 1 practice mark

Which statements are correct when comparing cash received today with future instalments?

A. Cash flows should be compared at a common valuation date

B. Future payments are discounted using an appropriate rate and timing

C. A larger nominal future total is necessarily more valuable than a smaller amount today

D. Risk, timing, and liquidity can matter to the comparison

Answer: (A, B and D)

Q49 · Multiple Selection Questions | 1 practice mark

Which statements correctly distinguish simple interest from compound interest?

A. Simple interest is calculated on the original principal in each period

B. Compound interest is calculated on the accumulated balance

C. For a positive rate and positive term, compound accumulation is generally at least as large as simple accumulation when the principal and rate basis match

D. Simple interest earns interest on previously accumulated interest

Answer: (A, B and C)

Q50 · Multiple Selection Questions | 1 practice mark

Which statements about rates and numerical accuracy are correct?

A. A nominal annual rate may differ from the effective annual rate when compounding occurs more than once a year

B. The effective annual rate includes the effect of within-year compounding

C. The nominal annual rate divided by m is the periodic rate, not the EAR

D. For a fixed positive nominal annual rate, increasing compounding frequency generally lowers the EAR

Answer: (A, B and C)

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